# Exact Ehrhart Numerators and Real-Rootedness for Multilayer Crown Order Polytopes: Tensor Networks and an Exact Theorem for $n = 3$

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**Abstract**

We study Ehrhart $h^*$-polynomials of order polytopes associated with the multilayer crown posets $C_{k,n}$. A proposed reduction to a one-dimensional transfer trace is valid in height two but fails already for $C_{3,3}$: at dilation one, the exact lattice-point count is 28, whereas the proposed trace gives 18. We express the count as a cylindrical two-dimensional tensor-network contraction and reconstruct the Ehrhart numerator exactly over $\mathbb{Z}$. For fixed dilation, the implemented row transfer gives an integer recurrence in the layer index by Cayley–Hamilton; its state-space bound depends on the dilation. Exact CRT contractions and rational Sturm or reduced $\Gamma$-Sturm certificates establish simple, strictly negative real roots in 19 cases, through $C_{8,5}$ of degree 32. Exact symbolic reconstruction finds finite-range recurrences for $n = 4$ and, for $n = 5$, rules out two bounded-degree order-four models; specialization at $t = 1$ has an exact scalar order-three recurrence with coefficients $(384, -10240, -32768)$. An exact six-state transfer proves the order-two recurrence for every $n = 3$ Ehrhart numerator. A phase argument then proves that all $2k$ roots are simple and lie in $(-7 - 4\sqrt{3}, -7 + 4\sqrt{3})$ for every $k \geq 2$. The remaining real-rootedness and uniform recurrence questions concern circumferences $n \geq 4$.

**Keywords:** Crown posets, Ehrhart polynomials, tensor networks, Sturm method, real-rootedness.

## 1 Introduction & Background

For a finite poset $P$, its order polytope is

$$\mathcal{O}(P) = \{x \in [0,1]^P : x_u \leq x_v \text{ whenever } u \leq_P v\}. \tag{1}$$

It is a full-dimensional lattice polytope of dimension $|P|$; its vertices are the characteristic vectors of upper sets of $P$ [2].

Fix integers $k \geq 2$ and $n \geq 3$. The poset $C_{k,n}$ has elements $x_{j,i}$ for $1 \leq j \leq k$ and $i \in \mathbb{Z}/n\mathbb{Z}$. Its cover relations are

$$x_{j,i} \prec x_{j+1,i}, \qquad x_{j,i} \prec x_{j+1,i+1} \quad (1 \leq j < k), \tag{2}$$

where the second index is read cyclically. The rank of $x_{j,i}$ is $j-1$, and $C_{k,n}$ has $d = kn$ elements.

Let

$$L_{k,n}(m) = |m\mathcal{O}(C_{k,n}) \cap \mathbb{Z}^d|, \qquad \sum_{m \geq 0} L_{k,n}(m)t^m = \frac{H_{k,n}(t)}{(1-t)^{d+1}}. \tag{3}$$

Thus $H_{k,n}(t) = h^*(\mathcal{O}(C_{k,n}), t)$ and $H_{k,n}(0) = 1$.

Lundström and Saud Maia Leite formulated the real-rootedness conjecture for the simple one-dimensional crown family in work originating in 2025 [1]; Cao and Fu subsequently proved

the height-two case $k = 2$ [4]. The present computational study addresses the multilayer families $C_{k,n}$ with $k \geq 3$. Its contributions are the failure of the proposed one-dimensional transfer trace beyond height two, a cylindrical two-dimensional tensor-network contraction, exact reconstruction of the Ehrhart numerators, and finite-data recurrence discovery. The recurrence and root-location statements for arbitrary parameters remain conjectural.

### 1.1 Methodological correction: the one-dimensional trace

For $N \geq 1$, define the matrix $K_N = (\min(a, b))_{1 \leq a,b \leq N}$. In height two, setting $N = m + 1$ gives the transfer identity

$$L_{2,n}(m) = \operatorname{tr}(K_N^n). \tag{4}$$

Indeed, fix the values on the bottom layer. For adjacent values $a_{i-1}, a_i$, the top-layer value at their common upper neighbor has $N - \max(a_{i-1}, a_i)$ choices. Replacing each bottom value $a_i$ by $N - a_i$ turns this factor into the matrix entry $\min(N - a_{i-1}, N - a_i)$. Summing around the cycle yields (4).

### 1.2 Failure of the proposed higher-layer extension

The height-two derivation sums out one layer and leaves pairwise interactions between neighboring values on a single cycle. With three or more layers, the intermediate variables remain coupled through two-dimensional local constraints; replacing those constraints by the proposed single transfer matrix does not preserve the count. The following exact example disproves that proposed extension.

**Proposition 1.1.** *For $C_{3,3}$ at dilation $m = 1$, the lattice-point count is*

$$L_{3,3}(1) = 28,$$

*whereas the proposed one-dimensional formula gives* $\operatorname{tr}(K_2^3) = 18$.

*Proof.* At $m = 1$, represent each layer by the subset of its three cyclic positions assigned value 1. A subset $A$ in one layer forces the next layer to contain $A \cup (A + 1)$. Counting the compatible layer choices gives $18 + 6 + 3 + 1 = 28$. In contrast,

$$K_2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \operatorname{tr}(K_2^3) = 18.$$

The two counts differ, so the proposed one-dimensional transfer-trace identity is not valid uniformly for $k \geq 3$. □

**Remark 1.2.** This counterexample rules out the stated transfer-trace reduction; it does not rule out other one-dimensional representations for special parameters, nor does it disprove Conjecture 5.1.

## 2 2D Boundary MPS Tensor Network Engine

### 2.1 Local factor and exact lattice-point count

For a fixed dilation $m \geq 0$, each coordinate takes values in $\{0, \ldots, m\}$. Define the local tensor

$$T_{a,b,c} = \begin{cases} 1, & c \geq \max(a, b), \\ 0, & \text{otherwise}, \end{cases} \qquad 0 \leq a, b, c \leq m. \tag{5}$$

Its three indices have link dimension $D = m + 1$. The indices $a, b$ encode adjacent values in one layer, and $c$ encodes the value above them. Thus the local factor enforces the required order constraint $x_{j+1,i} \geq \max(x_{j,i}, x_{j,i+1})$.

The complete cylindrical network is the factor-graph contraction

$$L_{k,n}(m) = \sum_{\substack{x_{j,i} \in \{0,\ldots,m\} \\ 1 \leq j \leq k,\ i \in \mathbb{Z}/n\mathbb{Z}}} \prod_{j=1}^{k-1} \prod_{i \in \mathbb{Z}/n\mathbb{Z}} T_{x_{j,i},\, x_{j,i+1},\, x_{j+1,i}}. \tag{6}$$

The cyclic index creates the periodic boundary in the circumferential direction. Each coordinate is summed once, and every factor in the product enforces one local cover constraint.

### 2.2 Boundary contraction and numerical modes

The tensor network is contracted by organizing its multilinear factors along the discrete cylindrical boundary and selecting an efficient hierarchy of index contractions. This is an exact contraction of the entire network: no boundary-rank compression or singular-value truncation is applied. Accordingly, "Boundary MPS" refers here to the boundary organization of the tensor contraction, not to an approximate matrix-product-state method.

The exact contraction is carried out over arbitrary-precision integers, so (6) involves no floating-point rounding. For spectral exploration, a separate numerical evaluation uses real or complex double-precision tensor arithmetic on the selected computing device. Numerical values are not used as integer certificates.

### 2.3 Layer transfer and its recurrence bound

For fixed $m$, let $X_{m,n} = \{0, \ldots, m\}^n$ be the set of row configurations and define the integer matrix

$$B_{m,n}(x, y) = \prod_{i \in \mathbb{Z}/n\mathbb{Z}} T_{x_i, x_{i+1}, y_i}, \qquad x, y \in X_{m,n}.$$

Summing over the intermediate rows in (6) gives

$$L_{k,n}(m) = \mathbf{1}^{\mathsf{T}} B_{m,n}^{k-1} \mathbf{1}.$$

The cyclic shift of coordinates commutes with $B_{m,n}$, so the transfer preserves the subspace fixed by $C_n$. Its rank is the number of cyclic orbits of $(m+1)$-colorings of an $n$-cycle,

$$M_{m,n} = \frac{1}{n} \sum_{r=0}^{n-1} (m+1)^{\gcd(n,r)}.$$

The orbit-sum basis makes the restricted transfer an integer matrix. If its characteristic polynomial is $z^r + \sum_{j=0}^{r-1} c_j z^j$, then Cayley–Hamilton yields

$$L_{k+r,n}(m) + \sum_{j=0}^{r-1} c_j L_{k+j,n}(m) = 0, \qquad r \leq M_{m,n}.$$

Indeed, multiply the characteristic-polynomial identity for $B_{m,n}|_{V_{m,n}^{C_n}}$ by $\mathbf{1}^{\mathsf{T}} B_{m,n}^{k-1}$ on the left and $\mathbf{1}$ on the right. The coefficient list $(c_0, \ldots, c_{r-1})$ lies in $\mathbb{Z}^r$. This proves an integer recurrence in $k$ for every fixed dilation. The bound and its coefficients depend on $m$, however, so this transfer does not by itself prove a recurrence for $H_{k,n}(t)$ with coefficients in $\mathbb{Z}[t]$ and a bound depending only on $n$. In particular, the implemented row transfer is not invariant under reflection: for

$n = 3, m = 1$, the transition $(1,0,0) \to (1,0,1)$ is allowed, but reflecting both rows gives the forbidden transition $(1,0,0) \to (1,1,0)$. Thus the cyclic orbit count of binary descent masks,

$$N_{C_n}(n) = \frac{1}{n} \sum_{d|n} \varphi(d) 2^{n/d},$$

is not the state-space dimension of this implemented operator. A separate P-partition transfer on descent masks would need an independent matrix-element identity with $H_{k,n}(t)$ before its Cayley–Hamilton bound could be applied. A uniform polynomial recurrence for the Ehrhart numerators therefore remains a separate question.

# 3 Exact Algebraic Reconstruction & Verification Results

## 3.1 Exact reconstruction of the Ehrhart numerator

Put $d = kn$. Ehrhart theory gives

$$L_{k,n}(m) = \sum_{j=0}^{d} h_j \binom{m+d-j}{d}, \qquad H_{k,n}(t) = \sum_{j=0}^{d} h_j t^j. \tag{7}$$

We evaluate (6) for $m = 0, 1, \ldots, d$ and solve the resulting triangular linear system by exact integer arithmetic. The coefficients above the structural degree $d - k$ vanish. No floating-point interpolation is used.

## 3.2 Exact results

Table 1 reports all 19 exact numerators obtained by CRT contraction and binomial-basis reconstruction. In every case the certificate counts as many negative roots as the degree and verifies simplicity, so all roots are real, strictly negative, and simple. For the even-degree palindromic cases $C_{5,5}$ through $C_{8,5}$, Sturm's method is applied exactly to the reduced $\Gamma$ polynomial. The coefficients are listed in descending order.

Table 1: Exact CRT dataset and root certificates; coefficient lists are descending.

| $k$ | $n$ | Degree | Negative roots | Method | Coefficients of $H_{k,n}(t)$ |
|---|---|---|---|---|---|
| 2 | 3 | 4 | 4/4 | CRT+Sturm | 1,11,24,11,1 |
| 3 | 3 | 6 | 6/6 | CRT+Sturm | 1,18,93,160,93,18,1 |
| 4 | 3 | 8 | 8/8 | CRT+Sturm | 1,25,211,739,1120,739,211,25,1 |
| 5 | 3 | 10 | 10/10 | CRT+Sturm | 1,32,378,2088,5757,8064,5757,2088,378,32,1 |
| 6 | 3 | 12 | 12/12 | CRT+Sturm | 1,39,594,4550,19005,44547,59136,44547,19005,4550,594,39,1 |
| 7 | 3 | 14 | 14/14 | CRT+Sturm | 1,46,859,8468,48263,165086,344061,439296,344061,165086,48263,8468,859,46,1 |
| 8 | 3 | 16 | 16/16 | CRT+Sturm | 1,53,1173,14185,103331,473631,1393927,2657795,3294720,2657795,1393927,473631,103331,14185,1173,53,1 |
| 3 | 4 | 9 | 9/9 | CRT+Sturm | 1,69,1204,7164,16906,16906,7164,1204,69,1 |
| 4 | 4 | 12 | 12/12 | CRT+Sturm | 1,100,3058,37164,204639,553200,767068,553200,204639,37164,3058,100,1 |
| 5 | 4 | 15 | 15/15 | CRT+Sturm | 1,131,5869,116355,1144241,6007079,17690249,30135467,30135467,17690249,6007079,1144241,116355,5869,131,1 |

*Continued on next page*

Table 1: Exact CRT dataset (continued).

| $k$ | $n$ | Degree | Negative roots | Method | Coefficients, descending |
|---|---|---|---|---|---|
| 6 | 4 | 18 | 18/18 | CRT+Sturm | 1,162,9641,273808,4129580,35320040,179517436,560710448,1100896046,1376570220,1100896046,560710448,179517436,35320040,4129580,273808,9641,162,1 |
| 7 | 4 | 21 | 21/21 | CRT+Sturm | 1,193,14374,539238,11319081,141272409,1094268784,5432987952,17714213382,38591690550,568061620 04,56806162004,38591690550,17714213382,5432987952,1094268784,141272409,11319081,539238,14374,193,1 |
| 2 | 5 | 8 | 8/8 | CRT+Sturm | 1,112,1983,9684,16120,9684,1983,112,1 |
| 3 | 5 | 12 | 12/12 | CRT+Sturm | 1,228,11096,175640,1128655,3314308,4717744,3314308,1128655,175640,11096,228,1 |
| 4 | 5 | 16 | 16/16 | CRT+Sturm | 1,349,32235,1090390,16336920,122193552,491902733,1115855805,1462319070,1115855805,491902733,122193552,16336920,1090390,32235,349,1 |
| 5 | 5 | 20 | 20/20 | CRT+$\Gamma$ | 1,470,67430,3915550,107636520,1560009346,12847278850,63317014410,193454540375,374892046080,466747051056,374892046080,193454540375,63317014410,12847278850,1560009346,107636520,3915550,67430,470,1 |
| 6 | 5 | 24 | 24/24 | CRT+$\Gamma$ | 1,591,117186,10199141,445096496,10693336966,151557685236,1333861396106,7570649136251,28485470286251,72457863450986,126239758904721,151779252595816,126239758904721,72457863450986,28485470286251,7570649136251,1333861396106,151557685236,10693336966,445096496,10199141,117186,591,1 |
| 7 | 5 | 28 | 28/28 | CRT+$\Gamma$ | 1,712,181578,21653944,1361039825,48903716880,1068736046140,14905774156760,137717986745175,867511817670400,3808653657683030,11848821044700960,26437684248135895,42655638313857400,50003623795120200,42655638313857400,26437684248135895,11848821044700960,3808653657683030,867511817670400,137717986745175,14905774156760,1068736046140,48903716880,1361039825,21653944,181578,712,1 |
| 8 | 5 | 32 | 32/32 | CRT+$\Gamma$ | 1,833,260611,40040390,3382607635,168934442784,5270888638227,107372156127054,1479487760034900,14185823530891700,96815201555014565,478811892910351720,1740289156615238005,4698922195864486730,9500786705237707205,14463987063668528165,16633278213336049990,14463987063668528165,9500786705237707205,4698922195864486730,1740289156615238005,478811892910351720,96815201555014565,14185823530891700,1479487760034900,107372156127054,5270888638227,168934442784,3382607635,40040390,260611,833,1 |

### 3.3 Sanity Checks & Data Verification

The 19 certified polynomials were checked against the exact coefficient lists in the JSON artifacts. Each satisfies

$$H_{k,n}(0) = 1, \qquad t^{k(n-1)} H_{k,n}(1/t) = H_{k,n}(t), \qquad \deg H_{k,n} = k(n-1).$$

The second identity is strict palindromicity (the Gorenstein reciprocity of these Ehrhart numerators); the degree agrees with Lemma 3.1. Independently, the JSON Sturm certificates record a negative-root count equal to the degree and a simplicity check for each of the 19 cases. Thus these checks apply to the listed finite dataset, not to the full infinite family.

### 3.4 Rational Sturm certificates

For a polynomial $H \in \mathbb{Z}[t]$, we form its Sturm sequence over $\mathbb{Q}$. Sign variations at $-\infty$ and at 0 give the exact number of distinct roots in $(-\infty, 0)$; this calculation involves rational polynomial arithmetic, not approximate root locations. Simplicity is checked independently by a gcd computation. For even-degree palindromic $H$, the optional reduced method uses $H(t) = (1+t)^d \Gamma(t/(1+t)^2)$ and Sturm's sequence of $\Gamma$. A full count of the degree in negative simple roots certifies the same conclusion for $H$.

Sturm sequences can also isolate individual roots: for rational endpoints $a < b$, the difference in sign variations at $a$ and $b$ counts the roots in $(a, b)$, and rational bisection can refine an interval until it contains exactly one root. The current JSON output records the exact polynomial and Sturm-sequence coefficients, endpoint sign lists, variation counts, root count, and simplicity/reciprocity checks; it does not report decimal approximations as certificates.

Exact rational Sturm certificates were computed for all 19 configurations: $n = 3$ for $k = 2, \ldots, 8$, $n = 4$ for $k = 3, \ldots, 7$, and $n = 5$ for $k = 2, \ldots, 8$. The four cases $C_{5,5}$ through $C_{8,5}$ use the exact reduced-$\Gamma$ method. Each JSON certificate records the degree check, endpoint signs, variation counts, root count, and simplicity check. The 19 exact certificates are structured in JSON format, named `certificado_k_n.json`, and available in the project's supplementary repository. [1]

### 3.5 Degree and reciprocity

**Lemma 3.1.** *For $k \geq 2$ and $n \geq 3$, $H_{k,n}(t)$ is monic of degree $kn - k$ and satisfies*

$$H_{k,n}(t) = t^{kn-k} H_{k,n}(1/t). \tag{8}$$

*Proof.* Let $P = C_{k,n}$ and $d = kn$. Lattice points of $m\mathcal{O}(P)$ are weakly order-preserving maps $P \to \{0, 1, \ldots, m\}$. Interior lattice points are maps $f : P \to \{1, \ldots, m-1\}$ that are strictly increasing on every cover relation.

Every maximal chain has $k$ elements. An interior lattice point therefore requires at least $k$ available integer values, so the smallest dilation with an interior lattice point is $m = k + 1$. At this dilation, each vertical chain $x_{1,i} \prec \cdots \prec x_{k,i}$ forces $f(x_{j,i}) = j$; these values satisfy all cover relations, giving exactly one interior lattice point. The codegree is consequently $k+1$. The standard degree-codegree relation for Ehrhart numerators gives $\deg H_{k,n} = d+1-(k+1) = d-k$, and the top coefficient equals the number of interior lattice points at the codegree, namely one. Thus $H_{k,n}$ is monic.

For reciprocity, let $\rho(x_{j,i}) = j - 1$. If $f$ is an interior lattice point of $m\mathcal{O}(P)$, then

$$g(x) = f(x) - \rho(x) - 1$$

is weakly order-preserving and takes values in $\{0, \ldots, m-k-1\}$. Conversely, every such weak map gives an interior lattice point by $f(x) = g(x) + \rho(x) + 1$. Hence the interior Ehrhart count is $E^\circ(m) = E(m-k-1)$ for $m \geq k+1$, and it is zero for smaller positive $m$. If $F(t) = \sum_{m \geq 0} E(m) t^m$, this says

$$\sum_{m \geq 1} E^\circ(m) t^m = t^{k+1} F(t).$$

[1] Execution scripts, CLI usage, and raw JSON certificate files are available in the open-source repository associated with this paper.

Ehrhart reciprocity gives $\sum_{m\geq 1} E^{\circ}(m)t^m = (-1)^{d+1}F(1/t)$. Substituting $F(t) = H_{k,n}(t)/(1-t)^{d+1}$ and simplifying yields (8). □

**Remark 3.2.** If $s = kn - k$ is odd, equation (8) forces $H_{k,n}(-1) = 0$. Under Conjecture 5.1, this root is simple and the other roots occur in reciprocal pairs.

**Proposition 3.3.** *For $k = 2$ and $n = 3$,*

$$H_{2,3}(t) = t^4 + 11t^3 + 24t^2 + 11t + 1,$$

*and its four roots are distinct and negative.*

*Proof.* Equation (4) gives $L_{2,3}(m) = 1, 18, 129, 571, 1884$ for $m = 0, 1, 2, 3, 4$. Substitution into (7) gives the displayed polynomial. Dividing by $t^2$ and setting $z = t + t^{-1}$ reduces its root equation to $z^2 + 11z + 22 = 0$. Its roots $(-11 \pm \sqrt{33})/2$ are distinct and less than $-2$, so each gives two distinct negative reciprocal roots of $t + t^{-1} = z$. □

# 4 Symbolic Recurrence Reconstruction

Solving exact coefficient systems over $\mathbb{Q}[t]$ against the tensor-network dataset gives the following finite-range propositions.

**Proposition 4.1** (Finite exact recurrence for $n = 3$)**.** *For the certified levels $k = 2, \ldots, 8$, the exact recurrence*

$$H_{k,3}(t) = (t^2 + 6t + 1)H_{k-1,3}(t) + (t^3 - 2t^2 + t)H_{k-2,3}(t) \tag{9}$$

*holds for every tested transition $k = 4, \ldots, 8$. The recurrence operator was reconstructed uniquely in the specified polynomial search space, and the coefficient system is overdetermined by the exact data. This proposition concerns only those certified transitions.*

*Proof.* The coefficients are determined over $\mathbb{Q}$ from the certified polynomials, then substituted into each transition using exact integer polynomial arithmetic. The identity holds at $k = 4, 5, 6, 7, 8$. The uniqueness claim is relative to the bounded-degree search used by the analyzer; it is not a uniqueness assertion among unrestricted operators of arbitrary degree. □

**Proposition 4.2** (Finite exact recurrence for $n = 4$)**.** *For the certified levels $k = 3, \ldots, 7$, the exact recurrence*

$$H_{k,4}(t) = P_4(t)H_{k-1,4}(t) + Q_4(t)H_{k-2,4}(t) + R_4(t)H_{k-3,4}(t) \tag{10}$$

*holds for the tested transitions $k = 6, 7$, where*

$$\begin{aligned} P_4(t) &= t^3 + 23t^2 + 23t + 1, \\ Q_4(t) &= 7t^5 - 28t^4 - 22t^3 - 28t^2 + 7t, \\ R_4(t) &= t^8 - 5t^7 + 9t^6 - 5t^5 \\ &\qquad - 5t^4 + 9t^3 - 5t^2 + t. \end{aligned}$$

*Within the specified degree bounds and reciprocal-polynomial ansatz, the exact coefficient system is overdetermined and has a unique integral solution. This proposition asserts the identity only at $k = 6, 7$.*

*Proof.* The three operators were reconstructed over $\mathbb{Q}$ using reciprocal spans $(3, 6, 9)$ and degree bounds $(4, 7, 10)$. Exact substitution into $H_{6,4}$ and $H_{7,4}$ gives zero residual in both transitions. The uniqueness statement is relative to this finite-dimensional ansatz. □

For $n = 5$, an unrestricted order-four search was also tested without symmetry or divisibility conditions. With degree bounds $(4, 12, 16, 20)$, the coefficient matrix has 56 columns and 99 equations; modulo $p = 2305843009213693951$, its rank and the augmented rank are 56 and 57. With expanded bounds $(6, 14, 20, 26)$, the corresponding values are 70 columns, 117 equations, and ranks 70 and 71. These modular ranks certify inconsistency over $\mathbb{Q}$: the coefficient matrix has full column rank, while the augmented matrix has strictly larger rank. Accordingly, neither bounded-degree model admits an exact order-four recurrence.

At $t = 1$, the exact sequence has minimum scalar recurrence order three:

$$H_{k,5}(1) = 384H_{k-1,5}(1) - 10240H_{k-2,5}(1) - 32768H_{k-3,5}(1), \qquad k = 5, 6, 7, 8. \tag{11}$$

The four certified transitions determine the three scalar coefficients uniquely and exclude scalar orders one and two.

The recurrence operators were uniquely reconstructed over $\mathbb{Q}[t]$ and certified on exact integer arithmetic. Here "uniquely" is relative to the bounded-degree search space for $n = 3$ and, for $n = 4$, to the stated degree bounds and reciprocal spans; it does not assert uniqueness without these restrictions.

All operator coefficients lie in $\mathbb{Z}[t]$ and are reciprocal with their indicated spans:

$$\begin{aligned} &P_3(t) = t^2 P_3(1/t), \quad Q_3(t) = t^4 Q_3(1/t), \\ &P_4(t) = t^3 P_4(1/t), \quad Q_4(t) = t^6 Q_4(1/t), \quad R_4(t) = t^9 R_4(1/t), \end{aligned}$$

where $P_3(t) = t^2 + 6t + 1$ and $Q_3(t) = t^3 - 2t^2 + t$. The operators have order $n - 1$ for the two circumferences investigated. The $n = 3$ identities were initially checked for transitions $k = 4, \ldots, 8$; Proposition 4.6 below proves the recurrence for every $k$. For $n = 4$, the two independent transitions $k = 6, 7$ determine a unique integral solution under reciprocal spans $(3, 6, 9)$ and degree bounds $(4, 7, 10)$.

**Remark 4.3.** The coefficients shown here are obtained by exact reconstruction and substitution into the certified polynomials. In particular, the $t^3$ coefficient of $Q_3$ and the $t^8$ coefficient of $R_4$ are both $+1$; changing either to $-1$ does not reproduce the dataset. These reconstruction checks are finite; the transfer proof below establishes the $n = 3$ identity for every $k$, while the $n = 4$ identity remains verified only on the stated range.

### 4.1 BKW spectral analysis for the order-three $n = 4$ recurrence

In this subsection, assume that the $n = 4$ recurrence (10) holds for every $k \geq 6$, with the exact initial polynomials $H_{3,4}, H_{4,4}, H_{5,4}$ from the certificates. The recurrence itself has so far been verified only at $k = 6, 7$. Let

$$\chi_4(\lambda; t) = \lambda^3 - P_4(t)\lambda^2 - Q_4(t)\lambda - R_4(t), \qquad R_4(t) = t(t-1)^6(t+1).$$

Its discriminant is

$$\Delta_4(t) = -4t\, q_4(t)\, q_6(t)^2,$$

where

$$q_4(t) = t^4 + 76t^3 - 282t^2 + 76t + 1, \qquad q_6(t) = t^6 + 7t^5 + 27t^4 + 58t^3 + 27t^2 + 7t + 1.$$

Writing $z = t + t^{-1}$ gives $q_4(t)/t^2 = z^2 + 76z - 284$. Its two negative roots are

$$\beta_\pm = \frac{z_0 \pm \sqrt{z_0^2 - 4}}{2}, \qquad z_0 = -38 - 24\sqrt{3}, \qquad \beta_+\beta_- = 1,$$

with $\beta_- \simeq -79.5566497221$ and $\beta_+ \simeq -0.01256965953$. Set $I_4 = (\beta_-, \beta_+)$.

**Theorem 4.4** (Negative-real BKW accumulation set for $n = 4$)**.** *For the sequence generated by the order-three recurrence from $H_{3,4}, H_{4,4}, H_{5,4}$, the set of accumulation points of its zeros on the negative real axis is exactly*

$$[\beta_-, \beta_+].$$

*For $t \in I_4$ away from the zeros of $q_6$, the two nonreal conjugate characteristic roots have equal modulus and strictly dominate the third root. At the zeros of $q_6$, this dominant pair coalesces into a repeated real root whose modulus still strictly exceeds that of the third root. For $t \in (-\infty, \beta_-) \cup (\beta_+, 0)$, the three characteristic roots are real and have pairwise distinct absolute values.*

*Proof.* For $t < 0$, the sign of $\Delta_4(t)$ is the sign of $q_4(t)$ away from the zeros of $q_6$. Thus, in $I_4$ there is one real root and a conjugate pair except at the zeros of $q_6$. To locate those exceptional points, put $z = t + t^{-1}$; then $q_6(t)/t^3 = z^3 + 7z^2 + 24z + 44$. This cubic is strictly increasing on $\mathbb{R}$, is negative at $z_0$ and positive at $-2$, so its unique root in $(z_0, -2)$ yields two reciprocal negative roots of $q_6$ in $I_4$. At each, the characteristic cubic has a repeated dominant real root. Outside the closure of $I_4$ there are three distinct real roots. A tie between the modulus of the conjugate pair and that of the real root would force $P_4(t)^3 R_4(t) + Q_4(t)^3 = 0$. Exact Sturm calculation shows that this polynomial has no negative zero. At $t = -1$,

$$\chi_4(\lambda; -1) = \lambda(\lambda^2 + 48),$$

so the conjugate pair is strictly dominant there as well. Continuity then gives the asserted strict dominance throughout $I_4$, including the isolated repeated-eigenvalue points where $q_6(t) = 0$.

Outside $I_4$, equality in absolute value between two distinct real roots would require them to be opposites, which implies $P_4Q_4 + R_4 = 0$. Its exact factorization is

$$P_4Q_4 + R_4 = 8t(t+1)(t^2 - 5t + 1)(t^4 + 20t^3 + 22t^2 + 20t + 1).$$

All its negative roots lie in $I_4$; exact root isolation at $t = -100$ and $t = -1/1000$ fixes the strict ordering on the two exterior components.

For clarity, the proposed condition $P_4^3 R_4 - Q_4^3 \neq 0$ throughout $I_4$ is false: that polynomial has one root in each of $(-50, -49)$ and $(-21/1000, -1/50)$, both contained in $I_4$. This condition is not needed for the dominance argument; the relevant no-crossing condition is $P_4^3 R_4 + Q_4^3 \neq 0$.

The numerator of the generating function for the recurrence is

$$N(\lambda, t) = \lambda^2 H_{3,4} + \lambda(H_{4,4} - P_4 H_{3,4}) + H_{5,4} - P_4 H_{4,4} - Q_4 H_{3,4}.$$

The exact resultant is

$$\mathrm{Res}_\lambda(\chi_4, N) = 16t^5(t-1)^{20}(t+1)^3 q_6(t)^2.$$

Consequently, away from the isolated repeated-eigenvalue points, the dominant-mode coefficients are nonzero on the interior of $I_4$ and the unique dominant coefficient is nonzero on the exterior. At $t = -1$, $N(\lambda, -1) = 1536\lambda$, so neither dominant coefficient vanishes. the Beraha–Kahane–Weiss theorem [3] therefore gives the stated accumulation set on $(-\infty, 0)$; the closure includes the two endpoints. □

**Remark 4.5.** The BKW accumulation theorem does not imply that every zero is real or simple for all sufficiently large $k$. Exact Sturm checks show that the recurrence-generated polynomials have $3k$ simple roots in $(-79, -1/79) \subset I_4$ for $k = 3, \ldots, 12$, but this finite range is not a proof for all large $k$. To make the assertion unconditional for the actual $H_{k,4}$, one must first prove the order-three recurrence for every $k$, and then establish a uniform real-zero count (for example, a sign-change argument producing $3k$ distinct zeros). No analogous recurrence or spectral analysis has been established here for $n \geq 5$.

### 4.2 Stage 1: an exact transfer and real-rootedness for $n = 3$

**Proposition 4.6** (Exact six-state transfer for $n = 3$)**.** *For the naturally labelled cylindrical poset, with $L(j, i) = 3j + i$ for $j \geq 0$ and $i \in \{0, 1, 2\}$, its Ehrhart numerator $H_{k,3}$ satisfies, for every $k \geq 2$,*

$$H_{k,3}(t) = P_3(t)H_{k-1,3}(t) + Q_3(t)H_{k-2,3}(t), \qquad P_3(t) = t^2 + 6t + 1, \quad Q_3(t) = t(t-1)^2,$$

*with $H_{0,3}(t) = 1$ and $H_{1,3}(t) = t^2 + 4t + 1$.*

*Proof.* For a naturally labelled poset, the Ehrhart $h^*$-polynomial is the descent enumerator of its linear extensions [2]. At a prefix of $3r$ elements, an order ideal is either $r$ complete layers or $r - 1$ complete layers followed by two partial layers of sizes two and one. Indeed, once a layer is not full, no later layer can be full; a partial layer of size two can be followed by at most one element, while one element cannot be followed by a nonempty layer. The only possible two-layer tails, written as bitmasks of vertices, are $(3, 2)$, $(5, 1)$, and $(6, 4)$. Remembering the last element gives the following six states:

| $s$ | partial-layer masks | last vertex |
|---|---|---|
| 1 | $(3, 2)$ | $(r, 1)$ |
| 2 | $(5, 1)$ | $(r, 0)$ |
| 3 | $(6, 4)$ | $(r, 2)$ |
| 4 | $r$ full layers | $(r - 1, 0)$ |
| 5 | $r$ full layers | $(r - 1, 1)$ |
| 6 | $r$ full layers | $(r - 1, 2)$ |

Here mask bit $i$ records whether vertex $i$ is present. Enumerating the legal ways to append the next three elements, with a factor $t$ for each new descent, gives the column-vector transfer

$$M(t) = \begin{pmatrix} t & t & 0 & t+1 & t+1 & t+1 \\ 0 & t & t & t+1 & t+1 & t+1 \\ t & 0 & t & t+1 & t+1 & t+1 \\ t^2 & 0 & t^2 & t^2+t & t^2+t & t^2+t \\ 0 & t^2 & t & 2t & 2t & 2t \\ t & t & 0 & t+1 & t+1 & t+1 \end{pmatrix}.$$

The exact initial prefix weights and terminal completion weights are

$$\mathbf{v} = \begin{pmatrix} (t+1)P_3 \\ (t+1)P_3 \\ (t+1)P_3 \\ t(t+1)P_3 \\ t(t+3)(3t+1) \\ (t+1)P_3 \end{pmatrix}, \qquad \alpha(t) = t(t+1), \quad \beta(t) = t^2 + 4t + 1, \quad \mathbf{g} = (\alpha, \alpha, \alpha, \beta, \beta, \beta).$$

Thus $H_{k,3}(t) = \mathbf{g}M(t)^{k-3}\mathbf{v}$ for $k \geq 3$. The transfer matrix and its boundary vectors are derived by exact enumeration in `TENSORES 2D/verify_n3_symbolic.py`.

Writing $\chi_3(\lambda; t) = \lambda^2 - P_3(t)\lambda - Q_3(t)$, direct symbolic calculation gives

$$\det(\lambda I - M) = \lambda^2(\lambda^2 - \lambda t + t^2)\chi_3(\lambda; t),$$

and the scalar resolvent is

$$\mathbf{g}(\lambda I - M)^{-1}\mathbf{v} = \frac{\lambda A(t) + B(t)}{\chi_3(\lambda; t)},$$

where

$$A(t) = t^6 + 18t^5 + 93t^4 + 160t^3 + 93t^2 + 18t + 1,$$
$$B(t) = t^7 + 9t^6 + 3t^5 - 26t^4 + 3t^3 + 9t^2 + t.$$

The numerator is coprime to $\chi_3$ over $\mathbb{Q}(t)$, so the scalar transfer sequence has minimal denominator $\chi_3$. Its coefficients therefore satisfy the asserted recurrence for $k \geq 5$. Direct exact enumeration gives $H_{0,3} = 1$, $H_{1,3} = t^2 + 4t + 1$ and verifies the remaining initial cases $k = 2, 3, 4$, proving the recurrence for every $k \geq 2$.

The six-state transfer is not $C_3$-equivariant: rotating the vertices does not preserve descent weights for the natural labeling. Accordingly, the proof uses the full six-state transfer and its scalar resolvent, not a quotient by cyclic symmetry. □

**Theorem 4.7** (Real-rootedness and negative simple roots for $n = 3$)**.** *For every $k \geq 2$, the actual Ehrhart numerator $H_{k,3}$ satisfies*

$$H_{k,3}(t) = P_3(t)H_{k-1,3}(t) + Q_3(t)H_{k-2,3}(t), \quad P_3(t) = t^2 + 6t + 1, \quad Q_3(t) = t(t-1)^2,$$

*with $H_{0,3}(t) = 1$ and $H_{1,3}(t) = t^2 + 4t + 1$. It is monic of degree $2k$ and has exactly $2k$ distinct, strictly negative real roots, all lying in*

$$I_3 = (-7 - 4\sqrt{3},\ -7 + 4\sqrt{3}) \subset (-\infty, 0).$$

*Proof.* Proposition 4.6 establishes the recurrence and initial values for the actual Ehrhart numerators. The characteristic discriminant factors as

$$\Delta_3(t) = P_3(t)^2 + 4Q_3(t) = (t+1)^2(t^2 + 14t + 1).$$

Writing $A(t) = -(t^2 + 14t + 1)$, we have $A(t) > 0$ on $I_3$ and

$$r(t) = (1-t)\sqrt{-t} > 0.$$

The characteristic roots are $r(t)e^{\pm i\theta(t)}$, where the continuous choice $\theta(t) \in (0, 2\pi)$ on $I_3$ is specified by

$$\cos\theta(t) = \frac{t^2 + 6t + 1}{2(1-t)\sqrt{-t}}, \qquad \sin\theta(t) = \frac{(t+1)\sqrt{A(t)}}{2(1-t)\sqrt{-t}}.$$

Since $2H_{1,3}(t) - P_3(t) = (t+1)^2$, solving for the Binet coefficients gives, away from the repeated-root point $t = -1$,

$$c_\pm(t) = \frac{1}{2} \mp \frac{i(t+1)}{2\sqrt{A(t)}}.$$

The resulting expression extends continuously through $t = -1$, where the characteristic root is repeated. The magnitude and argument of $c_+$ are

$$2|c_+(t)| = \sqrt{\frac{-12t}{A(t)}}, \qquad \phi(t) = \arg c_+(t) = \arctan\left(\frac{-(t+1)}{\sqrt{A(t)}}\right) \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right).$$

Consequently, throughout $I_3$,

$$H_{k,3}(t) = r(t)^k \sqrt{\frac{-12t}{A(t)}}\, \cos\big(\Psi_k(t)\big), \qquad \Psi_k(t) = k\theta(t) + \phi(t). \tag{12}$$

At $t = -1$, this identity holds by continuity (and gives $H_{k,3}(-1) = (-2)^k$).

Differentiating the phase yields

$$\Psi_k'(t) = \frac{-(k+1)t^2 + (10k+2)t - (k+1)}{2t(t-1)\sqrt{A(t)}}.$$

For $t \in I_3$, the denominator is positive, while the numerator is $-(k+1)(t^2+1) + (10k+2)t < 0$ for every $k \geq 1$. Hence $\Psi_k$ is strictly decreasing. At the two endpoints,

$$\lim_{t \to (-7-4\sqrt{3})^+} \Psi_k(t) = 2k\pi + \frac{\pi}{2}, \qquad \lim_{t \to (-7+4\sqrt{3})^-} \Psi_k(t) = -\frac{\pi}{2}.$$

Therefore, for each $m = 0, 1, \ldots, 2k-1$, there is exactly one $t \in I_3$ with $\Psi_k(t) = (m + \frac{1}{2})\pi$. Equation (12) makes each such $t$ a root. The prefactor is strictly positive and $\Psi_k'(t) \neq 0$, so differentiation of (12) shows each root is simple. These are $2k$ distinct roots. Finally, induction in the recurrence shows that $H_{k,3}$ is monic of degree $2k$: the term $P_3 H_{k-1,3}$ has degree $2k$ and leading coefficient one, while $Q_3 H_{k-2,3}$ has degree $2k-1$. Thus the displayed roots account for all roots. □

# 5 Real-Rootedness & Interlacing Implications

**Conjecture 5.1** (Higher-layer negative real-rootedness)**.** *For every $k \geq 3$ and $n \geq 4$, all zeros of $H_{k,n}(t)$ are negative real numbers and are simple.*

**Conjecture 5.2** (Cylindrical recurrence of order $n-1$)**.** *For each fixed circumference $n \geq 4$, there are polynomials $A_{1,n}(t), \ldots, A_{n-1,n}(t) \in \mathbb{Z}[t]$, independent of $k$, such that the recurrence holds for every $k \geq n$, where all lagged polynomials $H_{k-j,n}$ are defined:*

$$H_{k,n}(t) = \sum_{j=1}^{n-1} A_{j,n}(t) H_{k-j,n}(t).$$

The six-state transfer proves the order-two recurrence for every $H_{k,3}$. For $n = 4$, the order-three identity is verified only on the stated finite range. For $n = 5$, the order-four searches above exclude the two specified degree-bounded models, not every possible polynomial operator of arbitrary degree. The fixed-dilation transfer theorem proves recurrences for $L_{k,n}(m)$, but does not establish Conjecture 5.2 for $n \geq 4$. Theorem 4.7 settles real-rootedness unconditionally for $n = 3$; the remaining real-rootedness conjecture concerns $n \geq 4$.

## 5.1 Characteristic discriminant for the exact $n = 3$ recurrence

With the convention $H_{k,3} = P_3 H_{k-1,3} + Q_3 H_{k-2,3}$, the characteristic equation in the layer index is $\lambda^2 - P_3(t)\lambda - Q_3(t) = 0$. Its discriminant is therefore

$$\Delta_3(t) = P_3(t)^2 + 4Q_3(t) = t^4 + 16t^3 + 30t^2 + 16t + 1.$$

For $t < 0$, write $x = -t > 0$ and $z = x + x^{-1} \geq 2$. Then

$$\frac{\Delta_3(-x)}{x^2} = z^2 - 16z + 28 = (z-2)(z-14).$$

Consequently, $\Delta_3(t)$ vanishes at $z = 2$ and $z = 14$, is positive when $z > 14$, and is negative when $2 < z < 14$. Thus the characteristic multipliers are not real for every negative $t$; this discriminant does not by itself imply stability or real-rootedness of the polynomial sequence. The Sturm certificates establish the latter only for the individual cases in Table 1.

The alternative expression $P_3(t)^2 - 4(-t^3 - 2t^2 + t) = t^4 + 16t^3 + 46t^2 + 8t + 1$ is not positive for every $t < 0$: at $t = -4$ it equals $-63$. Moreover, the proposed $Q_3(t) = -t^3 - 2t^2 + t$ differs

from the verified operator $t^3-2t^2+t$ by $-2t^3$, so it does not reproduce the certified recurrences. Similarly, changing the leading coefficient of $R_4$ from $+1$ to $-1$ changes the recurrence residual by a nonzero multiple of $2t^8 H_{k-3,4}$. The coefficients in Section 4 are retained because they reproduce the exact data.

### 5.2 Interlacing status

The Sturm certificates prove that each listed polynomial has only simple negative real roots. They do not establish strict interlacing between different values of $k$. The degrees increase by $n-1$ in each fixed-$n$ family, so any interlacing claim requires a separately specified generalized interlacing criterion and an independent proof; no such general theorem is inferred from the current finite dataset.

## 6 Concluding Remarks & Future Work

The height-two transfer trace is justified by summing out a single layer, but the explicit $C_{3,3}$ count shows that the proposed one-dimensional extension does not preserve lattice-point counts for higher layers. The cylindrical tensor network retains the full local coupling, and its fixed-dilation row transfer proves integer recurrences for the lattice-point counts. For $n=3$, the exact six-state transfer proves an order-two recurrence for every Ehrhart numerator; the phase argument then gives $2k$ simple roots in $I_3$ for all $k \geq 2$. Exact CRT contraction and binomial-basis reconstruction yield reproducible integer polynomials; Sturm certificates verify simple negative real-rootedness for all 19 finite cases in Table 1, through $C_{8,5}$. Thus the remaining real-rootedness conjecture concerns $n \geq 4$. For $n=5$, two bounded-degree order-four systems are inconsistent, while the exact scalar specialization is (11). A uniform polynomial recurrence for $n \geq 4$ and higher-layer real-rootedness remain open. Future work includes constructing fixed-dimensional transfers for higher circumferences and proving root-location or generalized interlacing results uniformly in $k, n$.

The full-layer step raises the degree by $n-1$, while the finite tests of the transfer at $t=-s$ found neither total nonnegativity nor a diagonal sign conjugation to a nonnegative matrix. A vertex-by-vertex interpolation through order ideals, whose height profiles satisfy $h_i \leq h_{i-1 \pmod n}+1$, instead gives $C_{k,n}=P_0 \subset P_1 \subset \cdots \subset P_n = C_{k+1,n}$ and the exact decomposition $H_I(t)=\sum_{m\in\max(I)} H_I^{(m)}(t)$ by last maximal element. The maximal-deletion identity

$$H_I^{(m)}(t) = \sum_{w\in\max(I\setminus\{m\})} t^{\mathbf{1}[w>m]} H_{I\setminus\{m\}}^{(w)}(t)$$

was verified on all tested ideals. The directed interlacing sequence also held at every step of the five tested intermediate chains for $n \in \{3,4,5\}$; this finite evidence suggests a natural induction program for the conjecture at all $n \geq 4$, but does not establish such an induction. In particular, the exhaustive ideal test for $C_{3,5}$ had two exceptions to complete directed interlacing.

### 6.1 Data & Code Availability

The source code is in the sibling project directory `TENSORES 2D/`: the tensor-network engine and exact reconstruction modules are in `core/`, with command-line entry point `main.py`. The 19 JSON certificates accompanying Table 1 are distributed alongside this article. Sturm verification is carried out by `core/root_verifier.py`; recurrence and interlacing computations are carried out by `core/interlacing_analyzer.py`. The certificates record the polynomial coefficients and exact Sturm data needed to reproduce the finite checks reported here. The exact $n=3$ transfer proof is independently checked by `verify_n3_symbolic.py`; the $n=4$ spectral calculations are checked by `verify_n4_bkw.py`.

## References


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